

KUNAL ROY CHOUDHURY

[Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | +91-9954636583

EXPERIENCE

Software Engineer

March 2026 – Present

CollectEdge

Bengaluru, Karnataka

- Scaled event-driven microservices architecture in Rust to **50K+ daily messages** via horizontal partitioning across **5+ AWS SQS queues**, with backpressure-aware consumers, circuit-breaker controls, and idempotent processing; instrumented with **CloudWatch metrics** and structured logging for queue depth alerting and DLQ monitoring.
- Designed an AI-powered call analytics platform in React Native with **polyglot persistence (S3 + PostgreSQL)** and **CQRS** read/write separation, delivering async audio pipelines with ML inference for sentiment/disposition extraction and automatic interaction synthesis.
- Built a self-optimizing predictive dialer using weighted fair queueing, 5-tier priority strata, graph-based contact scoring to maximize agent utilization through real-time routing and ML-assisted dial timing.
- Architected [CredFix](#), a real-time credit risk engine running **18 parallel rule workers** on reactive streams with multi-bureau aggregation, achieving **sub-100ms** risk scoring via LRU caching and request batching.

Software Developer

April 2025 – Feb 2026

Indian Kanoon

Bengaluru, Karnataka

- Conceived, designed, and solo-developed [Prism](#), a production-grade legal platform scaling to **1.5 million users** from scratch leveraging **Django**, **PostgreSQL**, and **Redis**, while integrating **LLM** workflows across **10+ tools**.
- Engineered an LLM-powered RAG system leveraging **PostgreSQL text search vectors** with query expansion algorithms and context extraction, achieving **15% improvement** in legal search relevance, **92% query accuracy**, and **40% reduction** in latency through **Redis caching** and database optimization.
- Developed scalable OCR pipelines using **Tesseract** with batch processing for PDF, DOCX, and TXT formats via **Celery async processing**, processing **500+ legal documents** monthly.

Software Engineer Intern

Sept 2024 – April 2025

Chargebee

Chennai, Tamil Nadu

- Led development of a zero-downtime data migration framework leveraging **JOOQ schedulers** and an adaptive Vue.js validation system migrating 500K+ subscriptions while reducing query latency.

Software Engineer Intern

Jan 2024 – Sept 2024

AiDash

Bengaluru, Karnataka

- Designed a Spring Boot microservices framework using **Hexagonal Architecture** to decompose a **2M+ LOC monolith** (accelerating rollouts by **60%**), and optimized SQL pagination for retrieval of **100K+ records**.

PROJECTS

Gostman | *Go, React.js, Wails, REST APIs, GraphQL, WebSockets*

[live](#) | [code](#)

- Engineered a **privacy-first desktop API client** using **Wails (Go + React)** with local-only storage, request chaining, and unified REST/GraphQL/WebSocket support; achieved **<1s launch times** and **90% lower memory usage** than Electron-based alternatives.

Codebase Indexer | *Rust, Tree-sitter, Tantivy, ONNX, HNSW, RRF Fusion*

[code](#) | [article](#)

- Built a Cursor-style local semantic code search engine in **Rust** using Tree-sitter AST chunking, Tantivy BM25, and a custom HNSW vector index; implemented RRF fusion and embedded INT8 ONNX models locally to execute hybrid code retrieval without external dependencies.

TECHNICAL SKILLS

Generative AI/LLM: RAG systems, LangChain, vector embeddings, OCR (Tesseract)

Programming Languages: Python, Java, Golang, Rust, Dart, JavaScript, TypeScript, SQL

Backend & APIs: Django, Spring Boot, Express.js, RESTful APIs, GraphQL, async task processing (Celery)

Frontend & Mobile: React, React Native, Vue.js, Flutter, Redux, Zustand, Tailwind

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, PostgreSQL text search, ORMs

Cloud & DevOps: AWS (EC2, S3, SQS, RDS, CloudFront), Docker, Kubernetes, CI/CD, Git, GitHub Actions

EDUCATION

SRM Institute of Science and Technology

Kattankulathur, Tamil Nadu

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.24

Aug. 2020 – Sept 2024